

① Valovni vodnik iz dveh ravnih plošč
 |
 prenašanje EM valov v smeri z



$$\vec{E}(\vec{r}, t) = \vec{E}(\vec{\rho}) e^{i(kz - \omega t)} \quad \text{Nastavek}$$

↑
 koordinate v prečni ravnini (x, y ali r, φ)

Maxwell:

$$\left. \begin{aligned} \nabla \cdot \vec{E} &= 0 \\ \nabla \times \vec{E} &= -\mu_0 \frac{\partial \vec{H}}{\partial t} \\ \nabla \cdot \vec{H} &= 0 \\ \nabla \times \vec{H} &= \epsilon_0 \frac{\partial \vec{E}}{\partial t} \end{aligned} \right\} \begin{aligned} \nabla^2 \vec{E} - \frac{1}{c_0^2} \frac{\partial^2 \vec{E}}{\partial t^2} &= 0 \\ \nabla^2 \vec{H} - \frac{1}{c_0^2} \frac{\partial^2 \vec{H}}{\partial t^2} &= 0 \end{aligned} \quad \left| \begin{aligned} \text{za nastavek velja:} \\ \frac{\partial}{\partial z} &= ik \\ \frac{\partial}{\partial t} &= i\omega \end{aligned} \right.$$

$$\left(\nabla_{\vec{\rho}}^2 - k^2 + \frac{\omega^2}{c_0^2} \right) \vec{E}(\vec{\rho}) = 0$$

$$\left(\nabla_{\vec{\rho}}^2 + \left(\frac{\omega^2}{c_0^2} - k^2 \right) \right) \vec{E}(\vec{\rho}) = 0 \quad \text{enako za } \vec{H} \Rightarrow 6 \text{ komponent}$$

a) Pokaži, da je dovolj izračunati E_z in H_z , ostalo lahko izrazimo z njima

$$\begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} \times \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} = -\mu_0 \begin{bmatrix} \frac{\partial H_x}{\partial t} \\ \frac{\partial H_y}{\partial t} \\ \frac{\partial H_z}{\partial t} \end{bmatrix} = +\mu_0 i\omega \begin{bmatrix} H_x \\ H_y \\ H_z \end{bmatrix} = \begin{bmatrix} \frac{\partial E_z}{\partial y} - ikE_y \\ ikE_x - \frac{\partial E_z}{\partial x} \\ \frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} \end{bmatrix} \leftarrow$$

$$\begin{bmatrix} \frac{\partial H_z}{\partial y} - ikH_y \\ ikH_x - \frac{\partial H_z}{\partial x} \\ \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \end{bmatrix} = -\epsilon_0 i\omega \begin{bmatrix} E_x \\ E_y \\ E_z \end{bmatrix} \leftarrow$$

$$\left. \begin{aligned} \mu_0 i \omega H_x &= \frac{\partial E_z}{\partial y} - i k E_y \\ i k H_x - \frac{\partial H_z}{\partial x} &= -\epsilon_0 i \omega E_y \end{aligned} \right\} \Rightarrow E_y, H_x (E_z, H_z) \checkmark$$

E_x, H_y podobno

$$E_x = \frac{i}{k^2 - \frac{\omega^2}{c_0^2}} \left[-\omega \mu_0 \frac{\partial H_z}{\partial y} - k \frac{\partial E_z}{\partial x} \right] \quad H_x = \frac{i}{k^2 - \frac{\omega^2}{c_0^2}} \left[\omega \epsilon_0 \frac{\partial E_z}{\partial y} - k \frac{\partial H_z}{\partial x} \right]$$

$$E_y = \frac{i}{k^2 - \frac{\omega^2}{c_0^2}} \left[\omega \mu_0 \frac{\partial H_z}{\partial x} - k \frac{\partial E_z}{\partial y} \right] \quad H_y = \frac{i}{k^2 - \frac{\omega^2}{c_0^2}} \left[\omega \epsilon_0 \frac{\partial E_z}{\partial x} - k \frac{\partial H_z}{\partial y} \right]$$

Robni pogoji:

$$\nabla \times \vec{E} \xrightarrow{\int d\vec{s}} \int \vec{E} \cdot d\vec{l} \rightarrow E_{||} \Rightarrow \boxed{E_{||}|_0 = 0}$$

$$\nabla \cdot \vec{H} \xrightarrow{\int dV} \int \vec{H} \cdot d\vec{S} \rightarrow H_{\perp} \Rightarrow \boxed{H_{\perp}|_0 = 0}$$

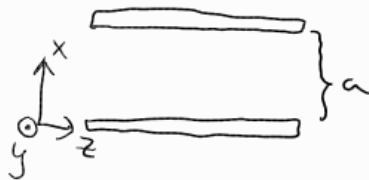
Če želimo valovanje, mora biti vsaj en izmed H_z in $E_z \neq 0$, predp. da en

Ločimo 2 primera:

1) $E_z \neq 0, H_z = 0$ TM (transverzalno magnetno)

2) $E_z = 0, H_z \neq 0$ TE (transverzalno električno)

b) TM način
 $E_z = ?$



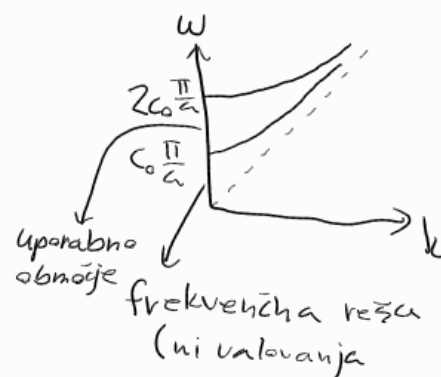
$$\frac{\partial}{\partial y} = 0 \quad \begin{aligned} H_z &= 0 \\ E_z &\neq 0 \end{aligned}$$

$$\Rightarrow H_x = 0, E_y = 0$$

$$\left[\frac{\partial^2}{\partial x^2} + k^2 \right] E_z = 0 \Rightarrow E_z = A \sin kx + B \cos kx$$

$$RP \Rightarrow k_n = \frac{n\pi}{a}, B = 0 \Rightarrow$$

$$\boxed{\omega = c_0 \sqrt{\left(\frac{n\pi}{a}\right)^2 + k^2}}$$



c) impedanca valovnega vodnika

$$Z = \frac{E_z}{H_{||}} = \frac{E_x}{H_y} = \frac{k}{\omega \epsilon_0} = \frac{\sqrt{\frac{\omega^2}{c_0^2} - \left(\frac{n\pi}{a}\right)^2}}{\omega \epsilon_0} = \frac{\cancel{\omega} \sqrt{1 - \left(\frac{n\pi c_0}{\omega a}\right)^2}}{\cancel{\omega} c_0 \epsilon_0} = \sqrt{\frac{\mu_0}{\epsilon_0}} \sqrt{1 - \left(\frac{n\pi c_0}{\omega a}\right)^2}$$

$$\frac{1}{c_0 \epsilon_0} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 376 \Omega = Z_0 \text{ Impedanca vakuma}$$

d) TE način

Začetek podoben, $H_z(x) = A \sin kx + B \cos kx$

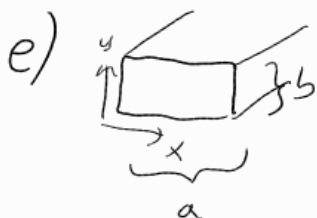
$$\text{RP: } H_x(0) = H_y(a) = 0$$

$$H_x = \frac{i}{k^2 - \frac{\omega^2}{c^2}} \left(\omega \epsilon_0 \frac{\partial E_z}{\partial y} - k \frac{\partial H_z}{\partial x} \right) \Rightarrow \frac{\partial H_z}{\partial x}(a) = \frac{\partial H_z}{\partial x}(0) = 0$$

$\downarrow$
0

$\downarrow$
0

$$\rightarrow A=0, \quad k = \frac{n\pi}{a}, \quad n=1, 2, \dots$$



$$(\nabla_{\perp}^2 + k^2) E_z(x, y) = 0$$

$$E_z = X(x)Y(y)$$

$$\underbrace{\frac{X''}{X}}_{-k_x^2} + \underbrace{\frac{Y''}{Y}}_{-k_y^2} + k^2 = 0 \rightarrow \begin{cases} X'' + k_x^2 X = 0 \rightarrow \sin k_x x, \quad k_x = \frac{n\pi}{a} \\ Y'' + k_y^2 Y = 0 \rightarrow \sin k_y y, \quad k_y = \frac{m\pi}{b} \end{cases}$$

$$k_x^2 + k_y^2 = k^2 = \frac{\omega^2}{c_0^2} + k^2$$

$$\text{RP: } \begin{matrix} E_{||}|_b = 0 \\ E_z|_0 \end{matrix}$$

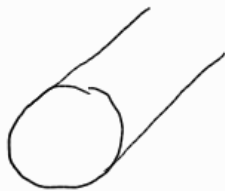
$$\rightarrow \omega = c_0 \sqrt{k^2 + \left(\frac{n\pi}{a}\right)^2 + \left(\frac{m\pi}{b}\right)^2}$$

Disperzijska relacija

2. Valjast val. vodnik

TM način

$$H_z = 0$$



$$(\nabla_{\perp}^2 + k^2) E_z(r, \varphi) = 0$$

↓

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2}{\partial \varphi^2}$$

$$E_z = R(r) \phi(\varphi)$$

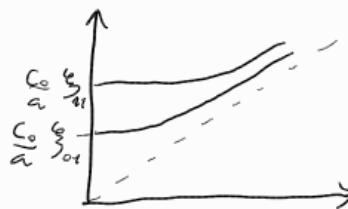
$$\phi'' + m^2 \phi = 0$$

$$\rightarrow \sin(m\varphi + \varphi_m)$$

$$R'' + \frac{1}{r} R' + (k^2 - m^2 \frac{R}{r^2}) R = 0 \rightarrow A_m J_m(kr) + B_m Y_m(kr)$$

$$RP: E_z(a, \varphi) = 0 \rightarrow ka = \xi_{mn}$$

$$\omega = c_0 \sqrt{k^2 + \left(\frac{\xi_{mn}}{a}\right)^2}$$



EM:

↓

$$E_z = \sum A_m J_m(kr) \sin(m\varphi + \varphi_m)$$

$$RP: \frac{\partial H_z}{\partial r}(a, \varphi) = 0 \rightarrow ka = \xi'_{mn} \leftarrow \text{hižke odvodov } J_m$$

$$\omega = c_0 \sqrt{k^2 + \left(\frac{\xi'_{mn}}{a}\right)^2}$$

Valovni vodnik s presekom dela kroga



TM

$$E_z = \sum A_m J_m(\chi r) \sin(m\varphi + \varphi_m)$$

$$\text{RP1: } J_m(\chi a) = 0 \rightarrow \chi = \frac{\chi_{mn}}{a}$$

$$\text{RP2: } E_z(r, 0) = 0 \rightarrow \varphi_m = 0$$

$$\text{RP3: } E_z(r, \pi/2) = 0 \rightarrow \sin(m \frac{\pi}{2}) = 0 \Rightarrow \underline{\underline{m \text{ sod}}}$$

4) TEM način v koaksialnem vodniku

$$E_z, H_z = 0$$



Za TEM način velja:

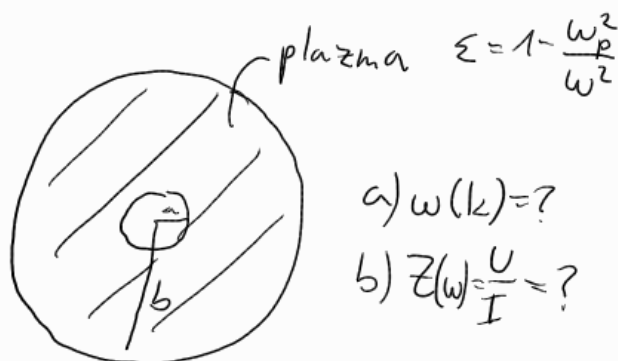
$$\left. \begin{aligned} \nabla \times \vec{E} = i\vec{k} \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t} = i\omega\mu_0 \vec{H} \\ \nabla \times \vec{H} = i\vec{k} \times \vec{H} = \epsilon_0 \frac{\partial \vec{E}}{\partial t} = -i\omega\epsilon_0 \vec{E} \end{aligned} \right\} \begin{aligned} \vec{H} &= \frac{\vec{k}}{\omega\mu_0} \times \vec{E} \\ \vec{E} &= -\frac{\vec{k}}{\omega\epsilon_0} \times \vec{H} \end{aligned}$$

$$\rightarrow \vec{E} = -\frac{\vec{k}}{\omega\epsilon_0} \times \left(\frac{\vec{k}}{\omega\mu_0} \times \vec{E} \right) = -\frac{c_0^2}{\omega^2} [\vec{k}(\vec{k} \cdot \vec{E}) - k^2 \vec{E}]$$

$$\vec{E} \left(1 - \frac{c_0^2 k^2}{\omega^2} \right) = 0 \rightarrow \boxed{\omega = c_0 k}$$

$$[\nabla_{\perp}^2 + (\frac{\omega^2}{c_0^2} - k^2)] \vec{E} = 0$$

$$\boxed{\begin{aligned} \nabla_{\perp}^2 \vec{E} &= 0 \\ \epsilon_0 /_0 &= 0 \end{aligned}} \quad \text{Laplace}$$



$$\text{a) } \omega(k) = ?$$

$$\text{b) } Z(\omega) = \frac{U}{I} = ?$$

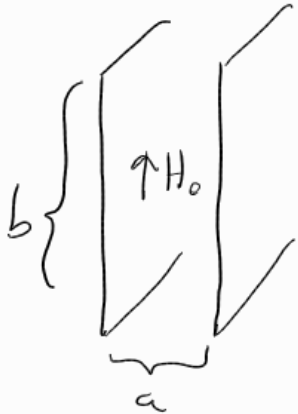
$$\text{a) } \omega = ck = \frac{1}{\sqrt{\epsilon}} c_0 k = \frac{c_0}{\sqrt{1 - \frac{\omega_p^2}{\omega^2}}} k$$

$$\omega^2 \left(1 - \frac{\omega_p^2}{\omega^2} \right) = c_0^2 k^2 \rightarrow \boxed{\omega = \sqrt{\omega_p^2 + c_0^2 k^2}}$$

$$\left. \begin{aligned} I &= 2\pi r H \\ V &= \epsilon_r \ln \frac{b}{a} \end{aligned} \right\} \text{od prej}$$

$$\begin{aligned} \rightarrow Z &= \frac{V}{I} = \frac{\ln \frac{b}{a}}{2\pi} \cancel{H} \frac{k}{\omega \epsilon_0 \epsilon} H = \frac{\ln \frac{b}{a}}{2\pi} \frac{\omega \sqrt{1 - \frac{\omega_p^2}{\omega^2}}}{\omega \epsilon_0 (1 - \frac{\omega_p^2}{\omega^2})} \epsilon_0 = \frac{\ln \frac{b}{a}}{2\pi \epsilon_0} \left(1 - \frac{\omega_p^2}{\omega^2}\right)^{-1/2} = \\ &= \boxed{Z_0 \left(1 - \frac{\omega_p^2}{\omega^2}\right)^{-1/2} \frac{\ln \frac{b}{a}}{2\pi}} \end{aligned}$$

⑤. Energijski tok



$$\langle \vec{P} \cdot d\vec{S} \rangle$$

$$\vec{P} = \vec{E} \times \vec{H}$$

$\boxed{TEM}$: $\langle \int \underbrace{EH}_{\text{vse pravokotno med sabo}} dS \rangle = \langle \int \frac{k}{\omega \epsilon_0} H^2 dS \rangle = \langle \frac{k}{\omega \epsilon_0} H^2 ab \rangle =$

$$= \frac{1}{2} \frac{k}{\omega \epsilon_0} H^2 ab =$$

$$= \frac{1}{2} \underline{\underline{Z_0 H^2 ab}}$$

$\boxed{TM}$: $(\vec{E} \times \vec{H}) \cdot d\vec{S} = \epsilon_x H_y dS$

$$\boxed{\langle \int \vec{P} \cdot d\vec{S} \rangle = \frac{1}{2} Z_0 H_0^2 \sqrt{1 - \frac{\omega_{min}^2}{\omega^2}} \frac{ab}{2}}$$